

End-to-End Max-Min Fairness Optimization for Two-Tier UAV Networks: A Differentiable Joint Trajectory Approach

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Abstract—The convergence of Sixth Generation (6G) Space-Air-Ground Integrated Networks (SAGIN) and Digital Twin (DT) systems demands ultra-reliable end-to-end data delivery from mobile ground users to the core network. A two-tier Unmanned Aerial Vehicle (UAV) architecture, with access UAVs and backhaul UAVs, provides a scalable solution, yet guaranteeing long-term fairness under user mobility remains an open challenge due to the coupled access-backhaul bottleneck. This paper formulates this problem as maximizing the worst-user cumulative throughput over a time horizon, and introduces Differentiable Joint Trajectory Optimization (DJTO), a fully differentiable framework that replaces hard max-min operations with smooth approximations, enabling gradient-based joint optimization of all UAV positions across all time slots. Comprehensive experiments against Decoupled Greedy (DG) and Model Predictive Control (MPC) baselines show that DJTO improves the poorest user’s cumulative throughput by 20.5% over DG and 20.4% over MPC, while spatiotemporal heatmap waterfalls visually confirm the superior end-to-end coverage and fairness.

Index Terms—6G, DT, differentiable optimization, end-to-end fairness, two-tier UAV networks.

I. INTRODUCTION

The Sixth Generation (6G) of wireless communication is expected to realize a fully connected world through Space-Air-Ground Integrated Networks (SAGIN), where aerial platforms such as Unmanned Aerial Vehicles (UAVs) play a pivotal role in extending coverage, enhancing capacity, and providing on-demand connectivity [1]. Concurrently, Digital Twin (DT) technology is emerging as a cornerstone for real-time monitoring and control of physical systems, requiring continuous and reliable data streams from distributed mobile sensors to the cloud [2]. In such contexts, a two-tier UAV relay architecture—where Lower-Tier UAVs (L1) serve as access points for ground users and Upper-Tier UAVs (L2) act as backhaul nodes to the core network—offers a flexible and cost-effective deployment model [3]. However, the dynamic nature of user mobility and the coupled relationship between access and backhaul links create significant challenges in guaranteeing end-to-end fairness over extended operational periods.

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Existing research on UAV trajectory optimization has largely focused on single-tier scenarios or instantaneous performance metrics. Many works optimize the positions of access UAVs under the assumption of ideal or static backhaul connectivity, thus ignoring the potential bottleneck caused by limited backhaul capacity [4], [5]. Others employ Model Predictive Control (MPC) or reinforcement learning to adjust trajectories in real time [6], [7], but these approaches typically optimize short-term objectives and may fail to ensure cumulative fairness over a long mission duration. Moreover, the hard max-min utility function commonly used in fairness-oriented designs is non-differentiable, making it difficult to jointly optimize over continuous spatial and temporal dimensions [8].

To bridge these gaps, this paper formulates the problem of long-term max-min fairness in a two-tier UAV network and proposes a solution with the following contributions:

- **Problem Formulation.** We model the end-to-end throughput of each user as the bottleneck of its access and backhaul links, and define the optimization objective as maximizing the minimum cumulative throughput over a finite time horizon, subject to UAV mobility constraints.
- **Differentiable Joint Trajectory Optimization (DJTO).** By replacing hard max and min operations with smooth Log-Sum-Exp (LSE) approximations [9], [10], we construct a fully differentiable loss function that allows gradient-based joint optimization of all UAV positions across all time slots. This framework, implemented in PyTorch, efficiently handles the coupled spatial-temporal variables.
- **Comprehensive Evaluation.** We compare DJTO with Decoupled Greedy (DG) and Model Predictive Control (MPC) baselines that jointly optimize both UAV tiers under the same speed penalty [11]. Our method improves the poorest user’s cumulative throughput by 20.5% over DG and 20.4% over MPC on average.
- **Visual Analytics.** We introduce waterfall-style spatiotemporal heatmaps that present the evolution of bottleneck rate across space and time, offering intuitive visual evidence of the dynamic coverage and fairness achieved by DJTO.

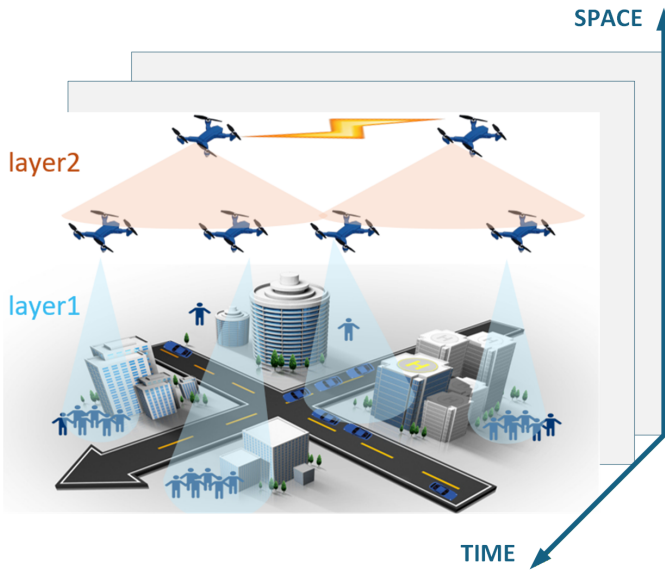


Fig. 1. Two-tier UAV network architecture: L1 access UAVs serve mobile users, L2 backhaul UAVs relay data to the core network.

The remainder of this paper is organized as follows. Section II presents the system model and the formal optimization problem. Section III describes the DJTO framework and the baseline methods. Section IV provides simulation settings and experimental results. Section V concludes the paper.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. Network Setup

We consider a two-tier UAV-assisted wireless network deployed over a square area of size $L \times L$ with $L = 100$ m. The network consists of $M = 50$ ground users, $N_1 = 4$ lower-tier access UAVs (L1), and $N_2 = 2$ upper-tier backhaul UAVs (L2). Time is discretized into $T = 20$ equal-length slots, indexed by $t = 0, 1, \dots, T-1$. The duration of each slot is normalized to one second. During each slot, all UAVs can adjust their horizontal positions subject to maximum speed constraints. The architecture is shown in Fig. 1.

The location of user u at slot t is denoted by $q_u(t) \in \mathbb{R}^2$. The positions of L1 UAV i and L2 UAV j at slot t are $p_i^1(t)$ and $p_j^2(t)$, respectively. Users move according to a hot-spot biased random walk model with three randomly placed hot-spot centers [11]. Each user is assigned to one hot spot and, at each time step, moves toward its assigned center with a pull of 0.15 times the displacement to the assigned center plus an independent Gaussian perturbation of scale 2.0 m. With probability $p_{\text{jump}} = 0.08$, a user may temporarily switch to a different hot spot to emulate group migration. All positions are confined within $[0, L]^2$.

B. Communication and Rate Model

We adopt a standard air-to-ground channel model based on logarithmic distance path loss and Shannon capacity [12], [13]. The channel uses large-scale path loss and AWGN (SNR);

small-scale fading, multi-user interference, and UAV altitude in the link distance are omitted. The path loss (in dB) at distance d is given by

$$PL(d) = P_{L0} + 10\alpha \log_{10}(d), \quad (1)$$

where P_{L0} is the reference path loss at 1 meter, and α is the path loss exponent. The received Signal-to-Noise Ratio (SNR) is

$$\text{SNR}(d) = \frac{P_t}{N_0 B_w \cdot 10^{PL(d)/10}}, \quad (2)$$

where P_t is the transmit power, N_0 is the noise Power Spectral Density (PSD), and B_w is the channel bandwidth. The achievable data rate (in bps) is then computed using the Shannon formula

$$f(d) = B_w \log_2(1 + \text{SNR}(d)). \quad (3)$$

For numerical stability, we convert the rate to Mbps. The access rate from user u to L1 UAV i at slot t is

$$a_{u,i}^t = f(\|q_u(t) - p_i^1(t)\|). \quad (4)$$

Similarly, the backhaul rate from L1 UAV i to L2 UAV j at slot t is

$$b_{i,j}^t = f(\|p_i^1(t) - p_j^2(t)\|). \quad (5)$$

Each L1 UAV selects the L2 UAV providing the highest backhaul rate, giving the effective backhaul rate of L1 UAV i as

$$b_i^t = \max_j b_{i,j}^t. \quad (6)$$

For user u served by L1 UAV i , the end-to-end rate is limited by the weaker of the access and backhaul links:

$$R_{u,i}^t = \min(a_{u,i}^t, b_i^t). \quad (7)$$

Here b_i^t is used as a per-user link bottleneck and does not impose a shared sum-rate constraint across users served by the same L1 UAV. Each user associates with the L1 UAV that maximizes this bottleneck rate, yielding

$$R_u^t = \max_i R_{u,i}^t = \max_i \min(a_{u,i}^t, \max_j b_{i,j}^t). \quad (8)$$

C. Problem Statement

Our objective is to maximize the cumulative throughput of the worst-performing user over the entire time horizon. The cumulative throughput of user u is

$$C_u = \sum_{t=0}^{T-1} R_u^t. \quad (9)$$

With unit slot duration, C_u is measured in Mbps.s. The optimization problem is formulated as

$$\max_{\{p_i^1(t)\}, \{p_j^2(t)\}} \min_u C_u, \quad (10)$$

subject to the following constraints for all i, j , and t :

$$\|p_i^1(t+1) - p_i^1(t)\| \leq v_{\max}^1 \quad (11)$$

$$\|p_j^2(t+1) - p_j^2(t)\| \leq v_{\max}^2 \quad (12)$$

$$0 \leq p_i^1(t), p_j^2(t) \leq L. \quad (13)$$

where $v_{\max}^1$ and $v_{\max}^2$ are the maximum travel distances per slot for L1 and L2 UAVs, respectively. Constraint (13) is enforced by projection after each update; (11) and (12) are handled by the speed penalty in Section III.

This problem is non-convex due to the nested max-min operations in (8) and the coupling of UAV positions across time slots. Traditional optimization methods are difficult to apply directly. In the next section, we develop a differentiable approximation that enables efficient gradient-based joint optimization.

III. DIFFERENTIABLE JOINT TRAJECTORY OPTIMIZATION

The optimization problem (10) is non-convex and combinatorial due to the nested max and min operations in the end-to-end rate (8). Moreover, the objective couples all UAV positions across all time slots, making it difficult to apply conventional iterative methods. To address this challenge, we propose DJTO, a framework that replaces all non-differentiable operations with smooth approximations, enabling gradient-based joint optimization of the entire trajectory set.

A. Smooth Approximations of Max and Min

The key idea is to approximate the hard max and min functions with their smooth counterparts using the LSE operator [9], [10]. For a set of values $x_1, \dots, x_K$ and a temperature parameter $\alpha > 0$, the smooth maximum is defined as

$$\text{smax}(x_1, \dots, x_K; \alpha) = \frac{1}{\alpha} \log \sum_{k=1}^K \exp(\alpha x_k), \quad (14)$$

and the smooth minimum is

$$\text{smin}(x_1, \dots, x_K; \alpha) = -\frac{1}{\alpha} \log \sum_{k=1}^K \exp(-\alpha x_k). \quad (15)$$

As $\alpha \rightarrow \infty$, these approximations converge to the exact max and min, respectively. For any $\mathbf{x} \in \mathbb{R}^K$, $\max_i x_i \leq \text{smax}(\mathbf{x}; \alpha) \leq \max_i x_i + \frac{\log K}{\alpha}$ (and symmetrically for smin). We use moderate α for stability and report the hard rates (8) after optimization.

Applying (14) and (15) to our rate model, we obtain the smooth backhaul rate of L1 UAV i :

$$b_i^{s,t} = \text{smax}_j(b_{i,j}^t; \alpha_{\text{bh}}), \quad (16)$$

where α_{bh} controls the smoothness of the backhaul selection. The smooth bottleneck rate for user u via L1 UAV i is

$$R_{u,i}^{s,t} = \text{smin}(a_{u,i}^t, b_i^{s,t}; \alpha_{\text{min}}), \quad (17)$$

and the smooth end-to-end rate for user u is

$$R_u^{s,t} = \text{smax}_i(R_{u,i}^{s,t}; \alpha_{\text{acc}}). \quad (18)$$

The smooth cumulative throughput of user u is then

$$C_u^s = \sum_{t=0}^{T-1} R_u^{s,t}. \quad (19)$$

Finally, the max-min fairness objective is approximated using the smooth minimum over users:

$$L_{\text{fair}} = -\text{smin}_u(C_u^s; \alpha_{\text{fair}}), \quad (20)$$

where the negative sign converts the maximization of the worst-user throughput into a minimization of the loss. Soft smax/smin is used in training; evaluation uses the hard association in (8).

B. Loss Function and Optimization

To incorporate UAV mobility constraints, we add a speed penalty term to the loss. The displacements of L1 and L2 UAVs between consecutive slots are given by

$$\Delta_{i,t}^1 = \|p_i^1(t+1) - p_i^1(t)\|, \quad (21)$$

$$\Delta_{j,t}^2 = \|p_j^2(t+1) - p_j^2(t)\|, \quad (22)$$

respectively. The speed penalty is defined as the sum of excess speeds beyond the allowed limits:

$$L_{\text{speed}} = \sum_{i=1}^{N_1} \sum_{t=0}^{T-2} \max(0, \Delta_{i,t}^1 - v_{\max}^1) + \sum_{j=1}^{N_2} \sum_{t=0}^{T-2} \max(0, \Delta_{j,t}^2 - v_{\max}^2). \quad (23)$$

The total loss function is

$$L = L_{\text{fair}} + \lambda L_{\text{speed}}, \quad (24)$$

where λ is a penalty coefficient that balances fairness optimization against mobility constraints.

All UAV positions across all time slots, i.e., $\{p_i^1(t)\}_{i=1, t=0}^{N_1, T-1}$ and $\{p_j^2(t)\}_{j=1, t=0}^{N_2, T-1}$, are treated as learnable parameters. We initialize these positions using K-means clustering of the user locations at the first time slot [14], which provides a reasonable starting point and ensures fair comparison with the baselines. The Adam optimizer [15] is then employed to minimize (24). After each gradient step, positions are projected onto $[0, L]^2$ to satisfy the spatial boundary constraint (13). The smoothed objective remains non-convex; Adam yields a local solution under the reported settings. The entire pipeline, from UAV positions to the loss value, is implemented in PyTorch, leveraging automatic differentiation for gradient computation.

C. Baseline Methods

We compare DJTO with DG and MPC. Both baselines jointly optimize L1 and L2 positions with the same speed penalty coefficient λ as DJTO, and both are evaluated with the same hard end-to-end association as DJTO after optimization.

DG: In each slot, both tiers are updated myopically from the previous positions. Using the same smooth approximations as DJTO, DG maximizes the instantaneous worst-user end-to-end rate in the current slot only, subject to the speed penalty, and then advances to the next slot. This captures a strong but short-sighted joint-tier baseline.

MPC: At each slot t , a prediction window of length H is formed [6]: future user locations are estimated by linear

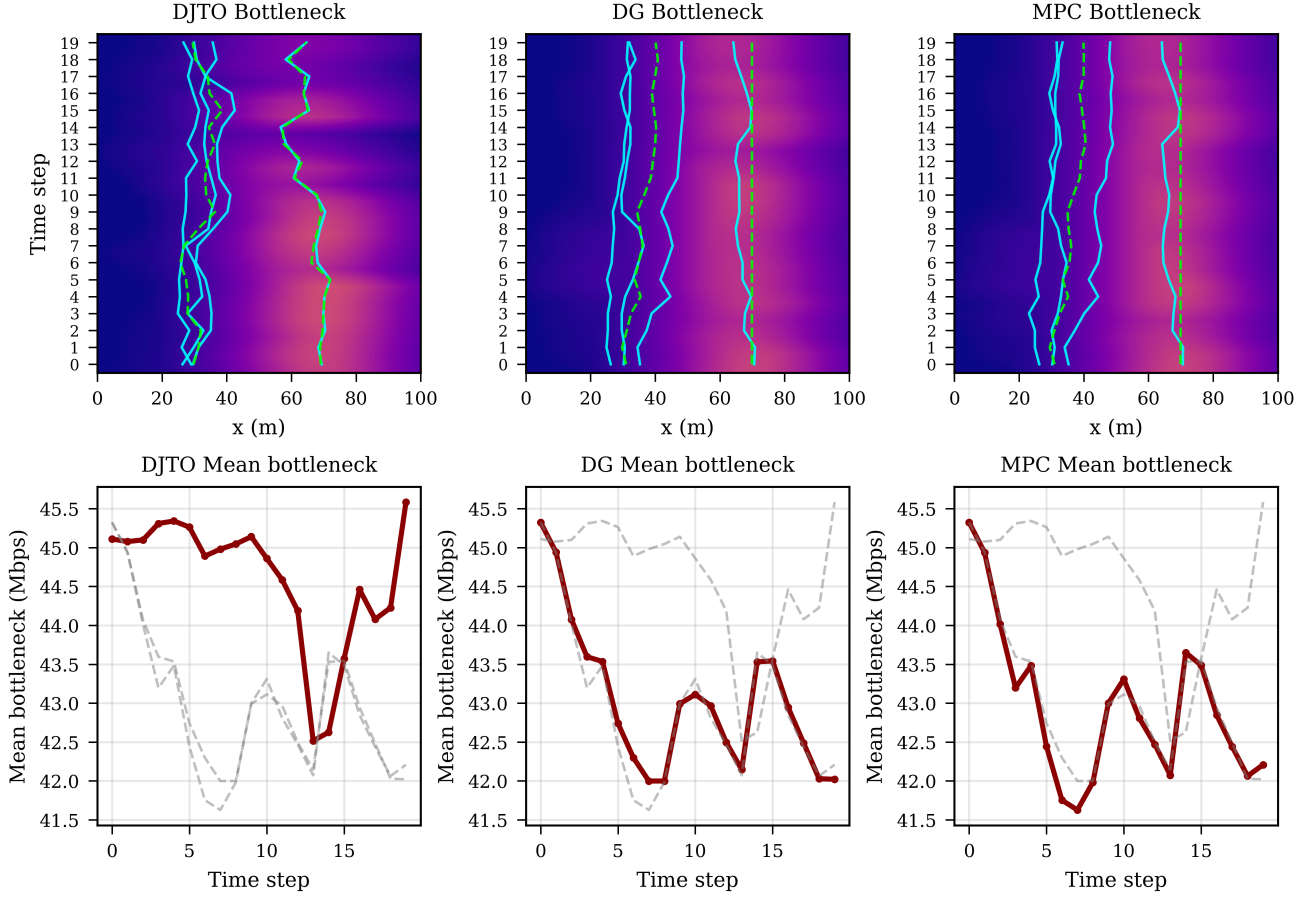


Fig. 2. Spatiotemporal bottleneck-rate waterfalls with UAV trajectory overlays. Top: bottleneck rate versus x and time. Bottom: mean bottleneck rate per slot. Solid and dashed lines show L1 and L2 trajectories for DJTO, DG, and MPC.

extrapolation from recent observations. L1 and L2 trajectories within the window are jointly optimized for the cumulative worst-user end-to-end rate under the same smooth loss and speed penalty. Only the first-step positions are applied, and the window rolls forward, mimicking predictive control with limited foresight.

IV. EXPERIMENTAL RESULTS

A. Simulation Settings

We evaluate DJTO and the two baselines in a simulated environment of area size $100 \times 100 \text{ m}^2$ with $M = 50$ mobile users and a two-tier UAV network consisting of $N_1 = 4$ access UAVs and $N_2 = 2$ backhaul UAVs over $T = 20$ time slots. The channel bandwidth is $B_w = 20 \text{ MHz}$, transmit power $P_t = 0.5 \text{ W}$ (27 dBm), noise PSD $N_0 = 10^{-13} \text{ W/Hz}$, path loss exponent $\alpha = 2.0$, and reference path loss $P_{L0} = 20 \text{ dB}$. The smoothness parameters are set to $\alpha_{\text{acc}} = \alpha_{\text{bh}} = \alpha_{\text{min}} = 10$ and $\alpha_{\text{fair}} = 5$. Maximum UAV speeds are $v_{\text{max}}^1 = 3 \text{ m/s}$ for L1 and $v_{\text{max}}^2 = 2 \text{ m/s}$ for L2, with penalty coefficient $\lambda = 0.05$. DJTO is trained for 400 iterations using the Adam optimizer with a learning rate of 0.2. DG and MPC each use 200 inner optimization steps per slot with the same λ . All UAVs are initialized via K-means clustering of the first-frame

TABLE I
SIMULATION PARAMETERS

parameter	value
Area size	$100 \times 100 \text{ m}^2$
Number of users M	50
L1 / L2 UAV count	4 / 2
Time slots T	20
Channel bandwidth B_w	20 MHz
Transmit power P_t	0.5 W (27 dBm)
Noise PSD N_0	10^{-13} W/Hz
Path loss exponent α	2.0
Reference path loss P_{L0} (1 m)	20 dB
Smoothness $\alpha_{\text{acc}}, \alpha_{\text{bh}}, \alpha_{\text{min}}$	10
Smoothness α_{fair}	5
Max L1 / L2 speed	3 / 2 m/s
Speed penalty λ	0.05
DJTO training steps	400
DJTO learning rate	0.2
Baseline inner steps	200
MPC prediction horizon H	3
Monte Carlo runs	20

user positions. Results are averaged over 20 independent user-mobility realizations. The key parameters are summarized in Table I.

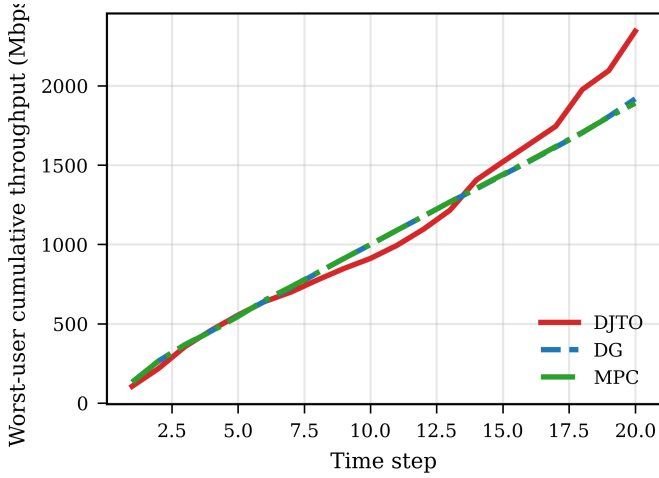


Fig. 3. Worst-user cumulative throughput (Mbps-s) versus time for DJTO, DG, and MPC.

TABLE II
WORST-USER CUMULATIVE THROUGHPUT $C_{\min}$ (MBPS-S)

Method	mean $\pm$ std
DJTO	2368.2 ± 238.2
DG	1969.3 ± 206.7
MPC	1971.4 ± 209.8

B. End-to-End Fairness Comparison

Table II reports the hard worst-user cumulative throughput $C_{\min}$ averaged over 20 independent realizations. On average, DJTO reaches 2368.2 ± 238.2 Mbps-s, which is 20.5% higher than DG (1969.3 ± 206.7) and 20.4% higher than MPC (1971.4 ± 209.8).

Fig. 3 shows the evolution of the worst-user cumulative throughput over the T time slots for a representative realization. DJTO (red solid curve) consistently outperforms both DG (blue dashed) and MPC (green dash-dot) from the early slots onward. At the final slot, the poorest user under DJTO accumulates 2343.47 Mbps-s, which is 22.2% higher than the 1917.29 Mbps-s achieved by DG and 24.0% higher than the 1890.10 Mbps-s of MPC. Furthermore, DJTO's advantage grows monotonically; the gap widens as more time slots are processed, confirming that the joint optimization of both tiers effectively prevents any user from being starved over the long term. These single-run curves are consistent with the average gains in Table II.

The mean cumulative throughput under DJTO reaches 2500.11 Mbps-s, compared with 2083.88 Mbps-s for DG and 2086.95 Mbps-s for MPC. The ratio of the minimum to the mean is 0.94 for DJTO, 0.92 for DG, and 0.91 for MPC. While all three methods benefit from jointly moving L1 and L2, DJTO still achieves both a higher floor and a higher average. In particular, MPC's worst user (1890.10 Mbps-s) remains farther from DJTO's floor than its mean would suggest, indicating that limited-horizon control does not fully close the fairness gap.

Fig. 4 presents the Cumulative Distribution Function (CDF) of the final per-user cumulative throughput. DJTO's CDF

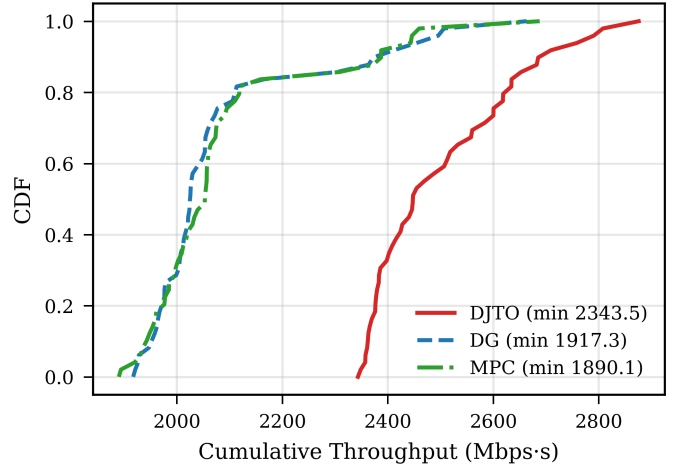


Fig. 4. CDF of per-user cumulative throughput after T slots.

curve (red solid) lies entirely to the right of those of DG (blue dashed) and MPC (green dash-dot), and rises steeply near the high-throughput region. This steep shape illustrates that all users under DJTO achieve similarly high throughput. By contrast, the CDF curves of DG and MPC exhibit longer left tails, indicating a group of users with substantially lower throughput.

Speed limits (11)–(12) are enforced softly through the penalty in (24). Under $\lambda = 0.05$, the mean per-step speed-violation rate is about 49% for DJTO and about 11%–12% for DG and MPC. Hard projection of the speed limits is left for future work.

C. Spatiotemporal Coverage Analysis

To understand how end-to-end coverage evolves over space and time, Fig. 2 visualizes the bottleneck rate via heatmap waterfalls and overlaid trajectory lines. Each column corresponds to one method; the heatmap shows the spatial distribution of bottleneck rate along the x -dimension (horizontal axis) versus time (vertical axis, from bottom to top).

Solid lines trace the L1 UAV centroid trajectories, while dashed lines trace the L2 UAV centroids.

DJTO's heatmap (left column) exhibits a consistently brighter appearance across all time slots, with high-rate regions persisting throughout the horizon. The bottom panel shows that DJTO's mean bottleneck rate remains comparatively stable over time. DG (middle column) also moves both tiers, yet its mean-rate curve fluctuates more strongly, reflecting the myopic per-slot updates that react to the current frame without planning over the full horizon. MPC (right column) improves temporal smoothness through the prediction window, but the limited foresight still yields a lower and less sustained bottleneck-rate profile than DJTO.

The overlaid trajectory lines provide additional insight into the coordination patterns. Under DJTO, the solid L1 centroid trajectory closely follows the spatial regions where users concentrate, actively bending toward the hotspots. Meanwhile, the

dashed L2 centroid adjusts its position dynamically over time, relocating to maintain strong backhaul coverage over the L1 cluster. This coordinated two-tier behavior is the direct result of the joint open-loop objective. In DG, the L1 trajectory undergoes more irregular jumps due to the per-slot independent optimization that disregards long-horizon continuity, while the L2 trajectory also moves but in a more reactive manner. Under MPC, the L1 trajectory is smoother than DG's owing to the predictive horizon, and L2 is likewise re-planned within each window; nevertheless, the truncated foresight limits how well the two tiers stay co-optimized over the full mission, which aligns with the $C_{\min}$ ranking in Table II.

These visualizations corroborate the quantitative results: DJTO's joint optimization of both tiers enables the access UAVs to actively track user mobility while the backhaul UAVs dynamically reposition to sustain high-quality backhaul, together achieving superior end-to-end fairness.

V. CONCLUSION

This paper addressed the problem of long-term max-min fairness in two-tier UAV networks with mobile users. We formulated the joint trajectory optimization problem with an end-to-end bottleneck rate model, developed the DJTO framework that replaces discrete max-min operations with smooth LSE approximations, and enabled efficient gradient-based optimization across all time slots. Simulations against DG and MPC show that DJTO improves the poorest user's cumulative throughput by 20.5% and 20.4%, respectively. Spatiotemporal heatmap waterfalls with overlaid UAV trajectories provided intuitive visual evidence of the coordinated two-tier behavior. Future work may include three-dimensional mobility, interference-aware channels, shared backhaul allocation, hard speed projection, and energy constraints.

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